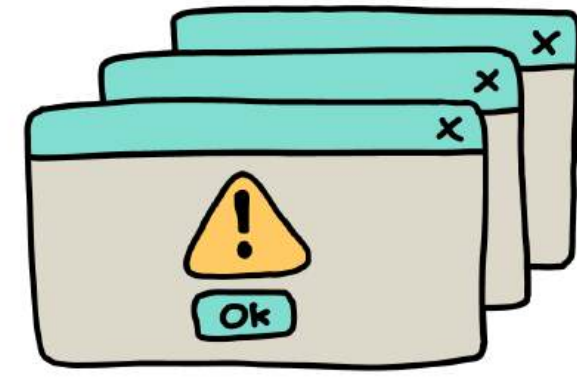




Inside the Biggest Outages: Hidden Failure Patterns Every SRE Should Know





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CKA, CKAD 📄

Public Speaker 🎤

Today's Agenda



01

The Pattern Problem: Why major outages keep happening

02

Case Studies: CrowdStrike, Cloudflare, AWS

03

Quick overview of what happened and
Deep dive into hidden technical patterns

04

Demo

05

Key Takeaways

The Pattern Problem - Why Major Outages Keep Happening

- **2024-2025:** Record number of planetary-scale outages
- **The Question:** Are systems getting worse? Or are we just seeing failures more clearly?
- **The Uncomfortable Truth:** Neither. We're repeating the same patterns.

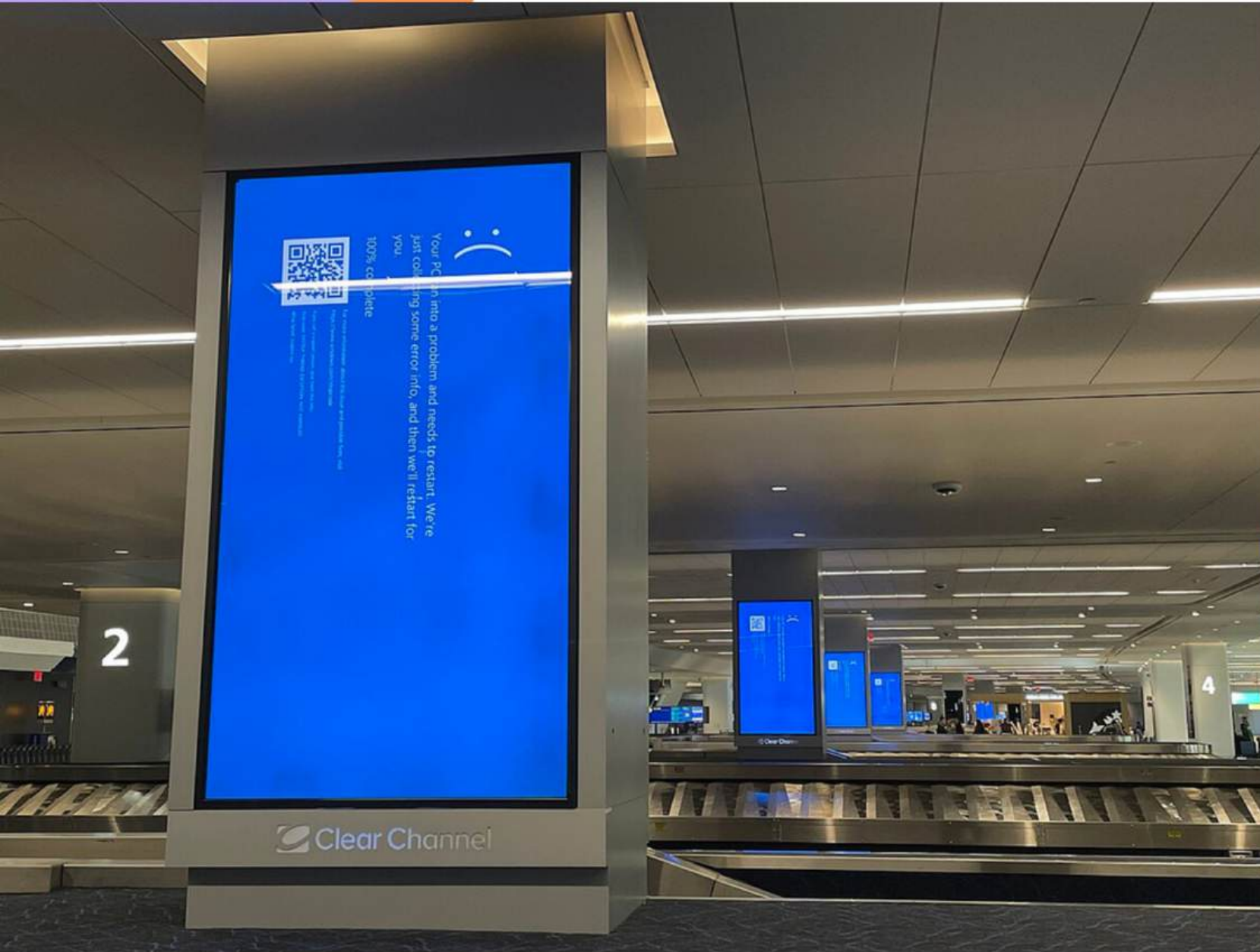


Why These Outages Matter

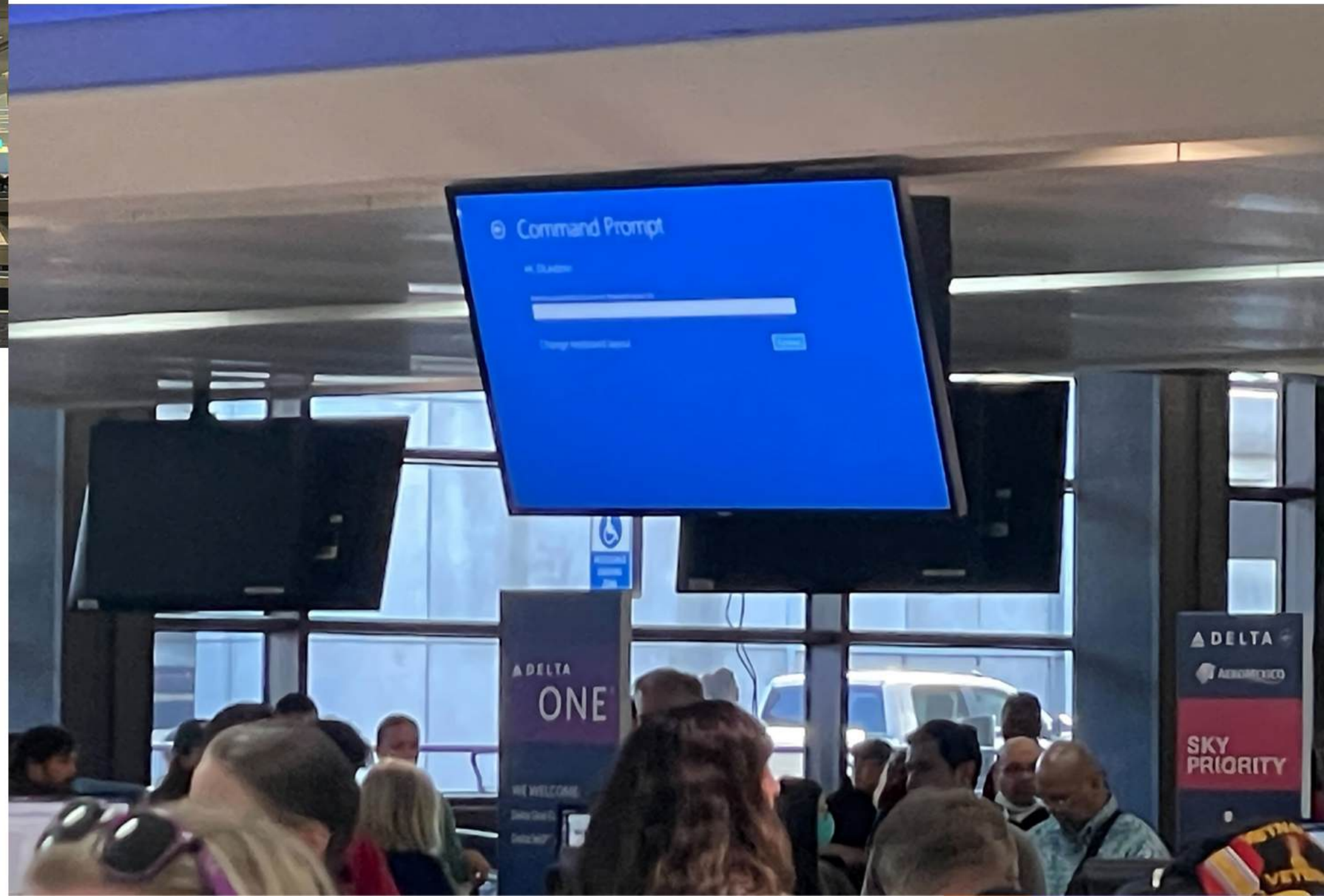
- Wide reach due to centralization
- Hidden fragility
- Cascading real-world impact
- Rise in frequency



Incident Overview - The Basics



CrowdStrike



CrowdStrike

- **Date & Time:** July 19, 2024, 04:09 UTC
- **What:** Faulty configuration update to CrowdStrike Falcon Sensor software
- **Who:** ~8.5 million Windows PCs and servers worldwide
- **Scale:** One of the largest IT outages in history
- **Single Root Cause:** Logic error in "Channel File 291" configuration update

CrowdStrike

**ONE DOES NOT SIMPLY
UPGRADE**

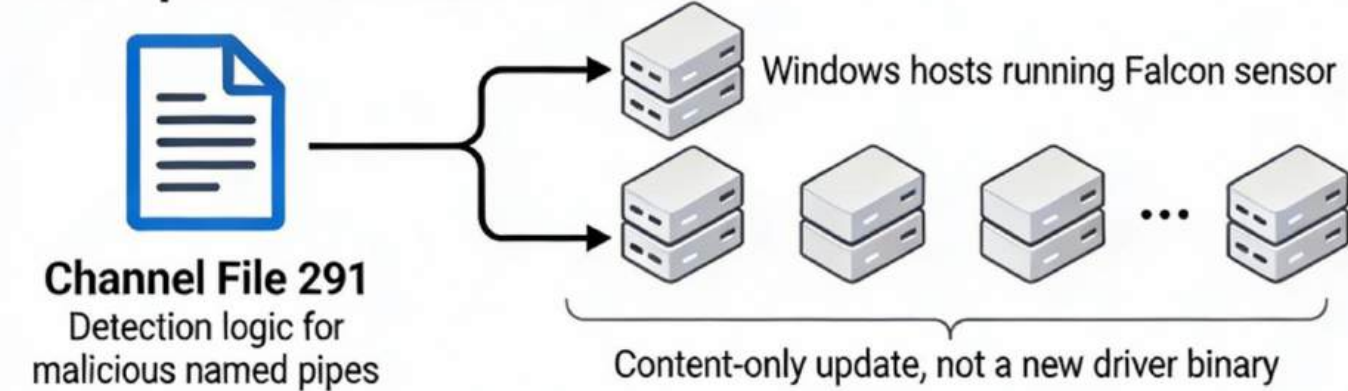
**ONE CAUSES THE LONGEST
UNPLANNED OUTAGE DURING A
PLANNED "NO OUTAGE" UPGRADE**

Root Cause - The Technical Details

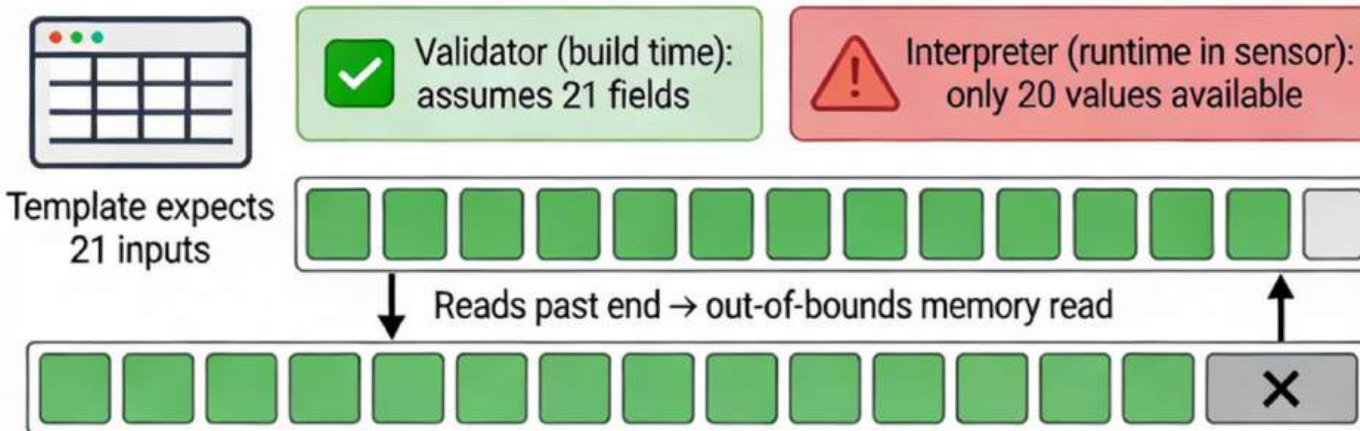
- **The Update:** "Channel File 291" - a configuration file for detecting malicious named pipes
- **The Bug:** Out-of-bounds memory read caused by parameter mismatch
- **The Mechanism:** Logic error in the file interacted with kernel-level code
- **Why Critical:** Falcon runs at the Windows kernel level (ring 0 - highest privilege)



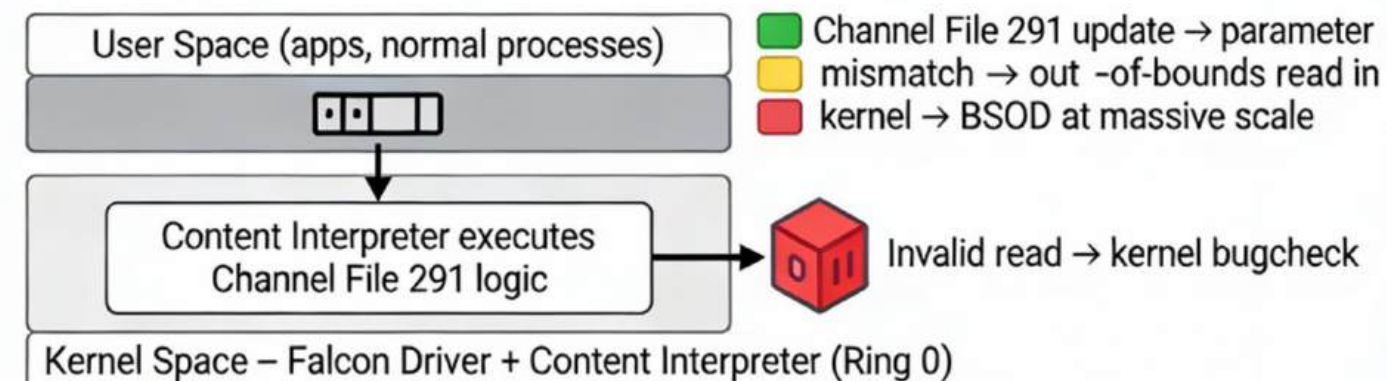
The Update: Channel File 291



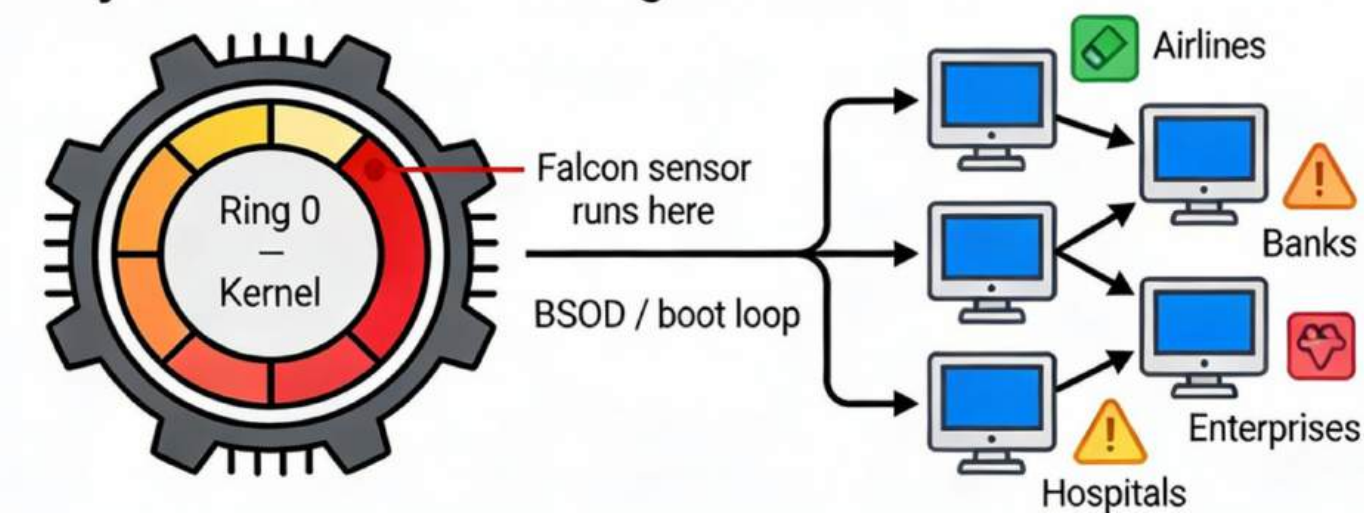
The Bug: Parameter Mismatch → Out-of-Bounds Read



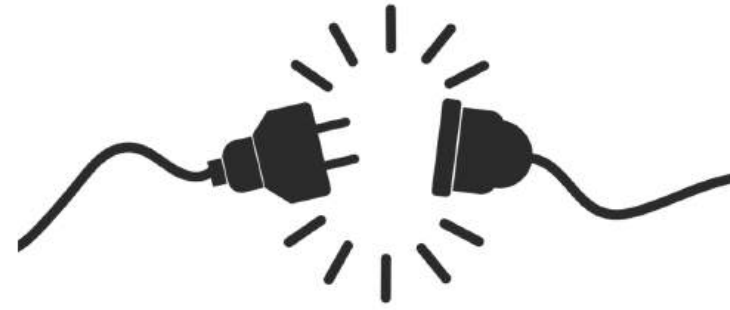
The Mechanism: Content Logic + Kernel-Level Code



Why Critical: Kernel Privilege → Global BSOD



The Cascade Effect - What Broke



- **Airlines:** AI, Indigo, Spicejet, Delta, Southwest airlines.
- **Hospitals:** Emergency departments unable to access patient records
- **Banks:** ATMs offline, transactions blocked
- **Government:** Courts, agencies unable to function
- **Media & Broadcasting:** TV stations went dark
- **Businesses:** Microsoft 365, Gmail, Slack all appeared down due to client devices



Employees when there is power cut or internet issues in middle of work



Key Lessons Learned

- Configuration is as dangerous as code
- Kernel-level access = zero margin for error
- Manual recovery at scale is a nightmare
- Rollout velocity matters for blast radius



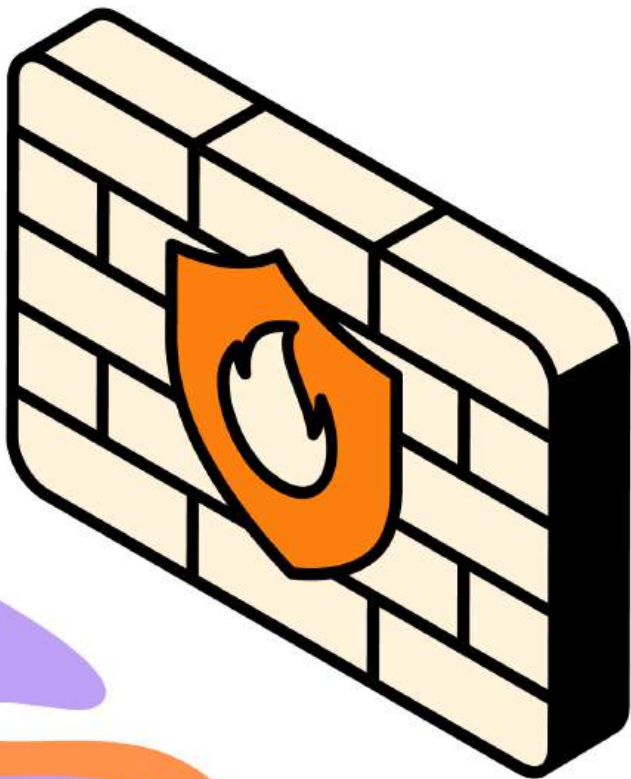
How to Prevent Similar Incidents

- Treat configuration with code-level rigor
- Implement multi-stage progressive rollouts



How to Prevent Similar Incidents

- Comprehensive validation testing
 - negative test cases: malformed data, null values
 - Chaos testing
- Build safety mechanisms into critical components
 - Safe mode boot options
 - Circuit breakers
- Maintain rollback mechanisms





Incident Overview – The Basics

Internal server error Error code 500

Visit cloudflare.com for more information.

2025-11-18 11:35:32 UTC



You
Browser
Working



London
Cloudflare
Error



sancloudflare.com
Host
Working

What happened?

There is an internal server error on Cloudflare's network.

What can I do?

Please try again in a few minutes.

- > Cloudflare failure
- > You want read some details about this malfunction
- > Portal with article is not working because it is using Cloudflare



CloudFlare

The Incidents - Timeline Overview

- **November 18, 2025 at 11:20 UTC:** Bot Management feature file bug causes cascading failures
- **Duration:** ~5 hours 38 minutes (11:20 UTC → 17:06 UTC)
- **December 5, 2025 at 08:47 UTC:** Configuration change for security mitigation causes brief outage
- **Duration:** ~25 minutes (08:47 UTC → 09:12 UTC)
- **Key Context:** Both incidents involved deployment propagation across a global edge network
- **Scale:** Millions of websites affected, 28% HTTP traffic, Zoom/LinkedIn/Discord down

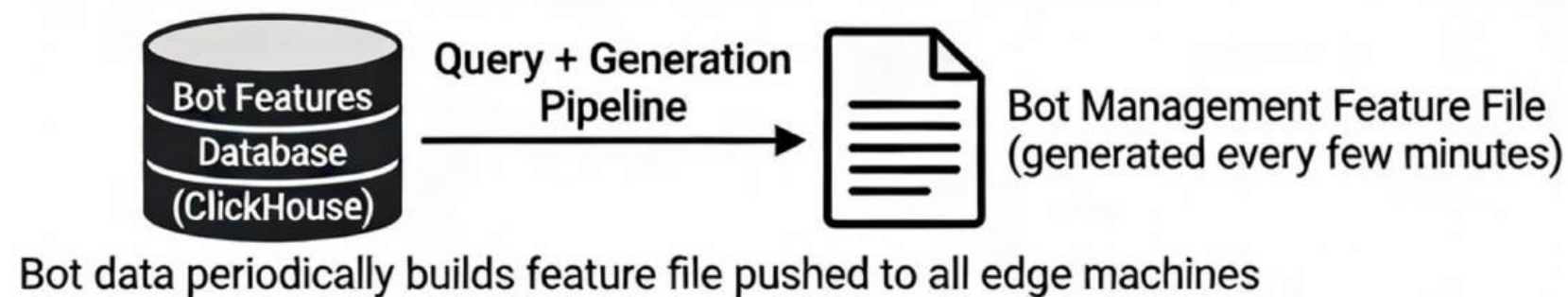


November 18 Incident – Root Cause

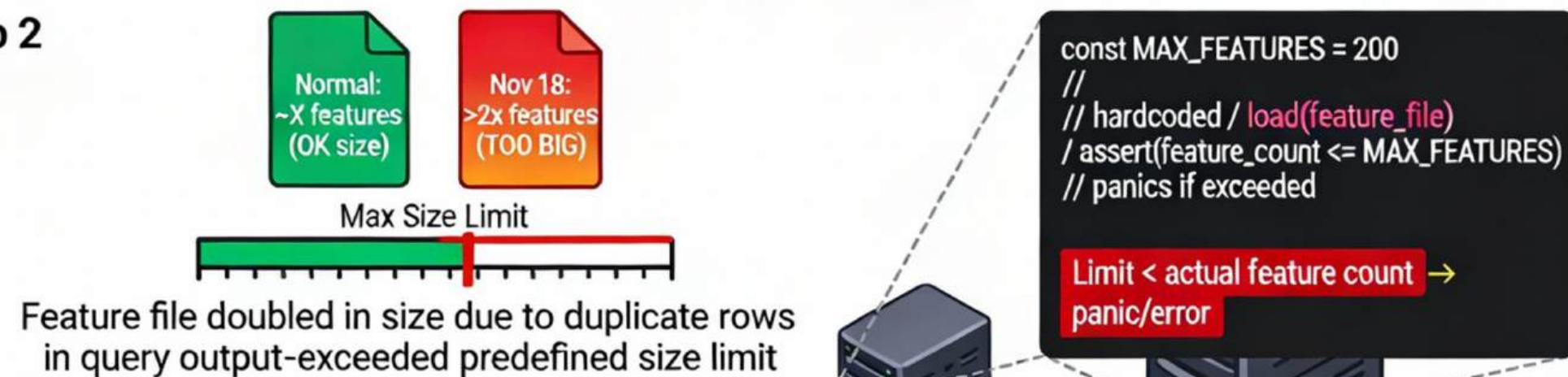
- **What Failed:** Bot Management feature file generation logic
- **The Bug:** Feature file exceeded predefined size limits in downstream systems
- **Root Cause:** Software had a hardcoded size limit that was below the actual file size; no validation before propagation
- **Why It Mattered:** The oversized file triggered failures across many Cloudflare services
- **Pattern:** Similar to CrowdStrike; a parameter mismatch between two components
- **Propagation:** The file propagated to the entire network before being detected



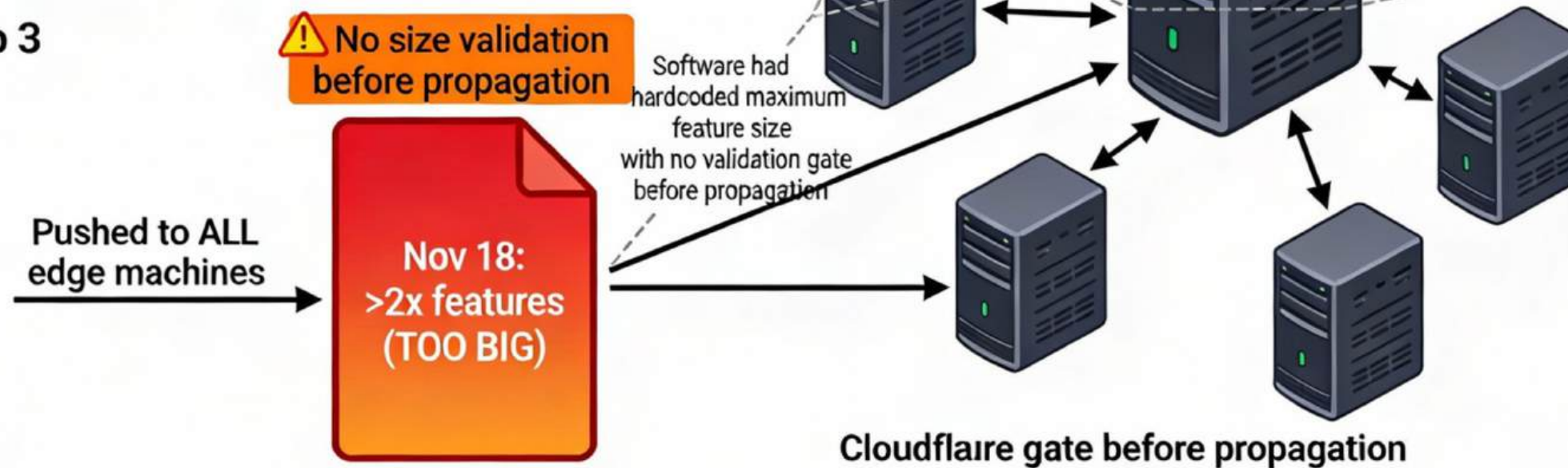
Step 1



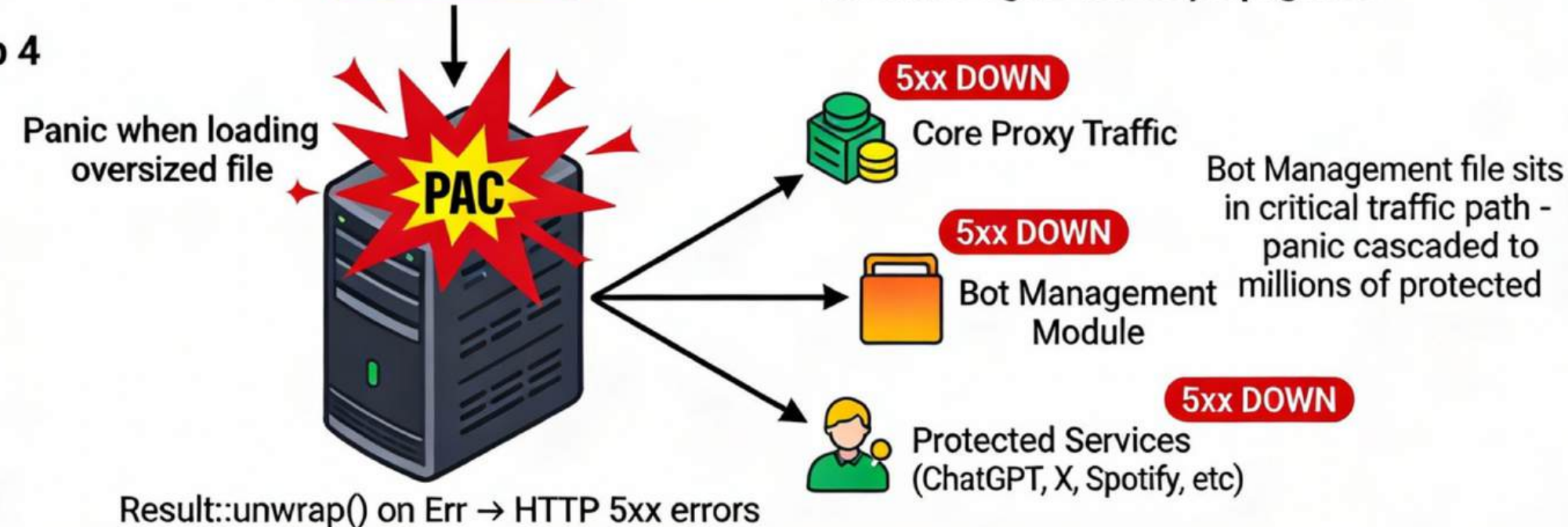
Step 2



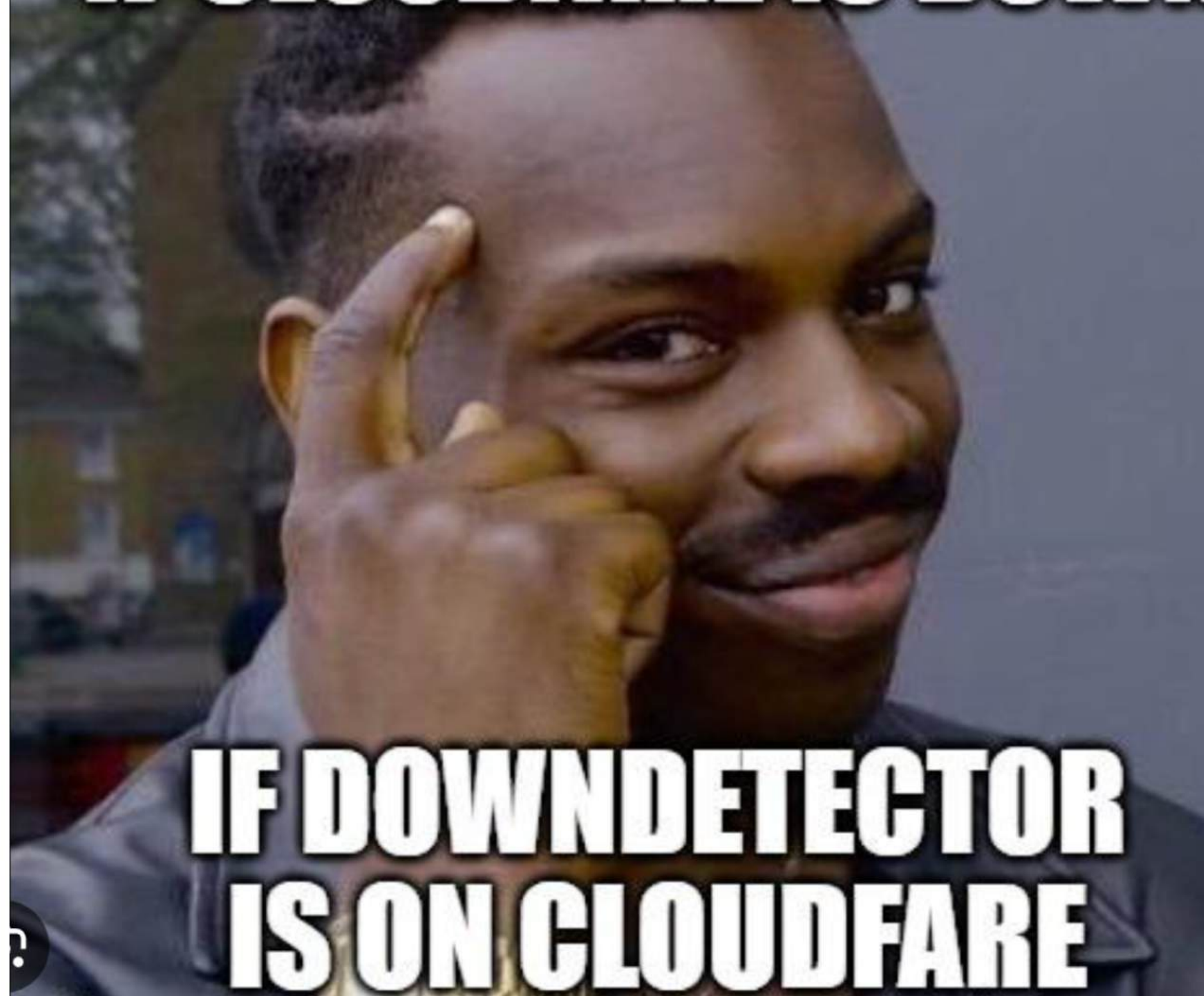
Step 3



Step 4



**YOU CANT TELL
IF CLOUDFARE IS DOWN**



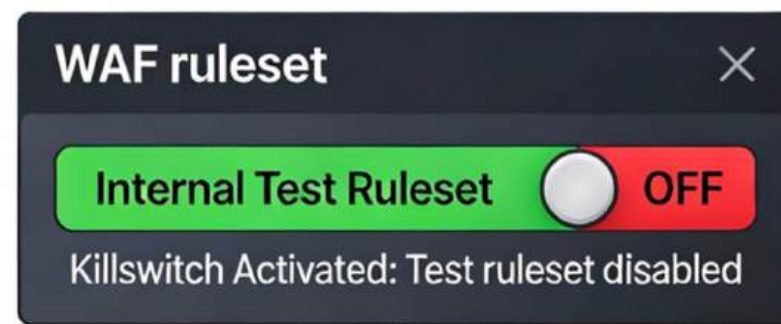
**IF DOWNDETECTOR
IS ON CLOUDFARE**

December 5 Incident - Security Patch Gone Wrong

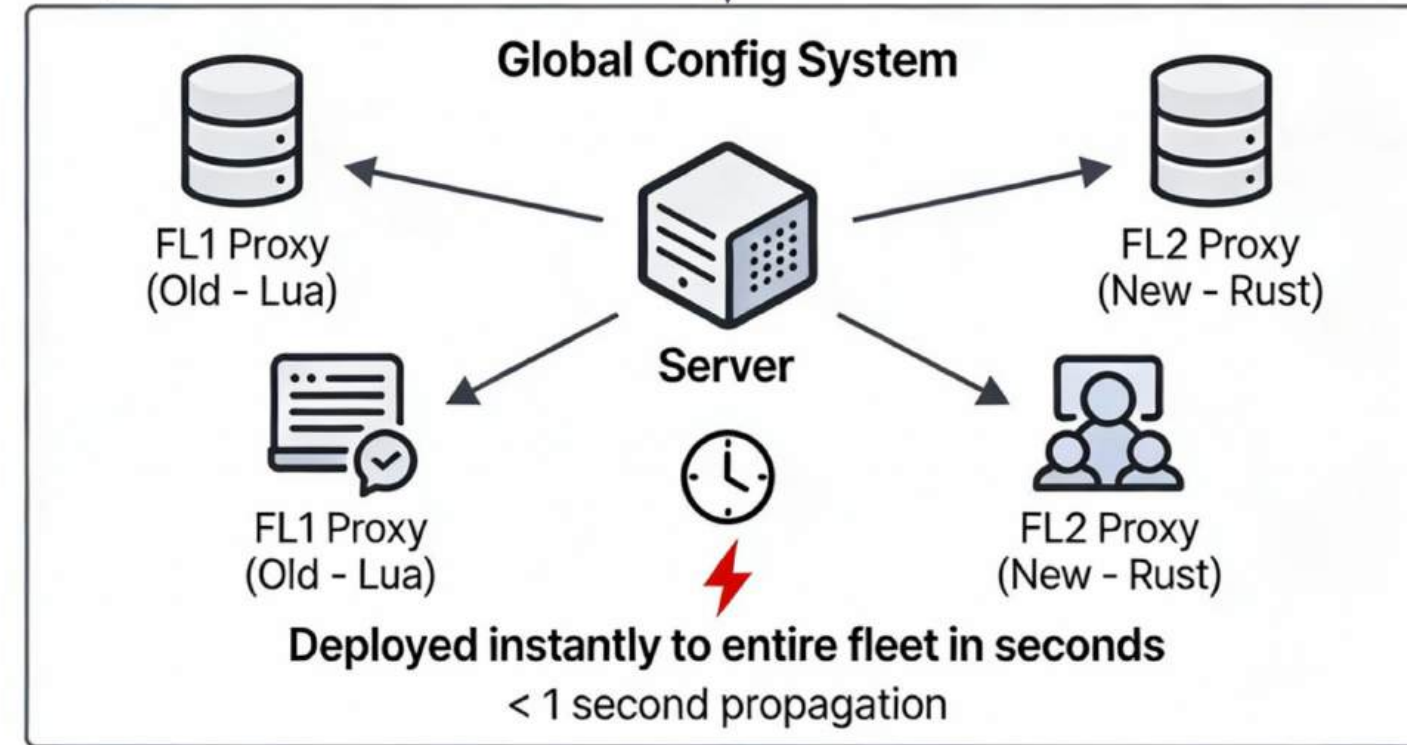
- **Context:** Recent industry-wide React Server Components vulnerability discovered
- **Config change:** Disabled a WAF test ruleset using a global "killswitch"
 - The killswitch skipped evaluation of a rule whose action was "execute".
- **Applied globally:** Pushed to entire fleet in seconds
- **Old FL1 code:** Had a hidden bug; assumed "execute" object would always exist
- **The bug:** Tried to read from non-existent object → Lua error → HTTP 500
- **The Impact:** Caused errors for ~28% of HTTP traffic
- **Resolution:** Reverted within 25 minutes once detected



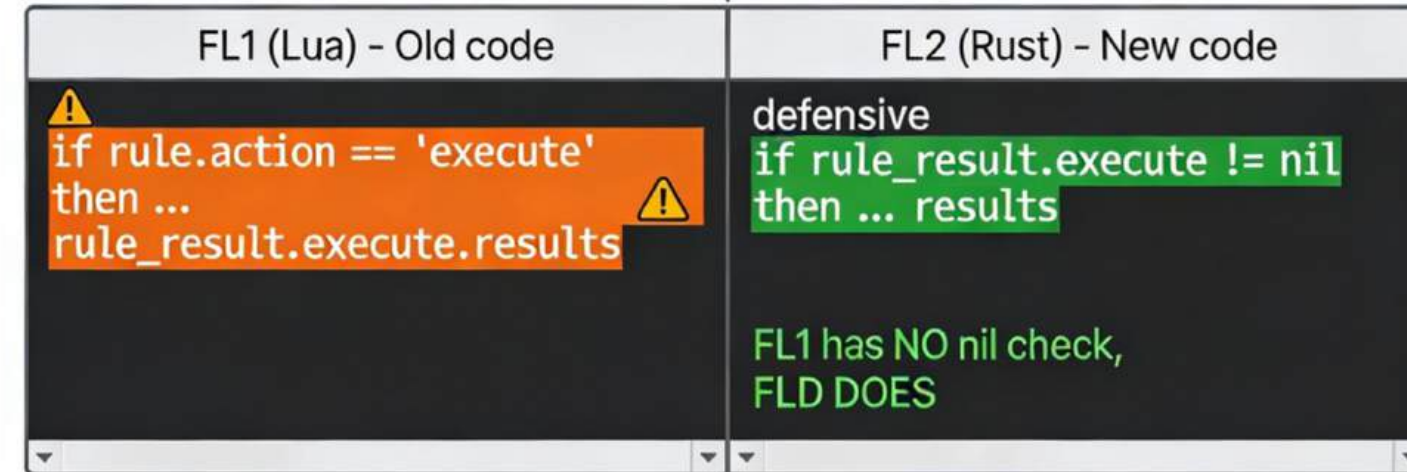
Step 1



Step 2



Step 3



Step 4



Services Impacted - Global & India Context

- **Globally Affected:** X (Twitter), ChatGPT, Spotify, Canva, Discord, LinkedIn, Zoom, Multiple platforms like- WhatsApp Web, banking portals, Paytm, Netflix, major news sites
- **Financial Services:** Online banking, payment platforms experienced degradation
- **E-commerce:** Amazon, Flipkart, and other retail sites behind Cloudflare CDN
- **User Experience:** Users saw 5xx errors, timeouts, or very slow page loads; not complete unavailability for most



How to Prevent Similar Incidents

- **Enhanced pre-deployment validation**

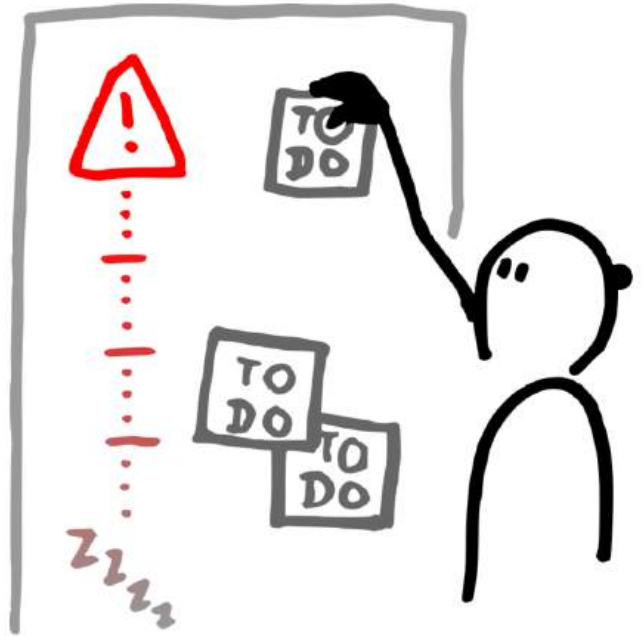
- Build production-equivalent staging environments for network changes
- Implement "shadow mode" testing where changes run in parallel without affecting traffic

- **Gradual rollout for infrastructure changes**

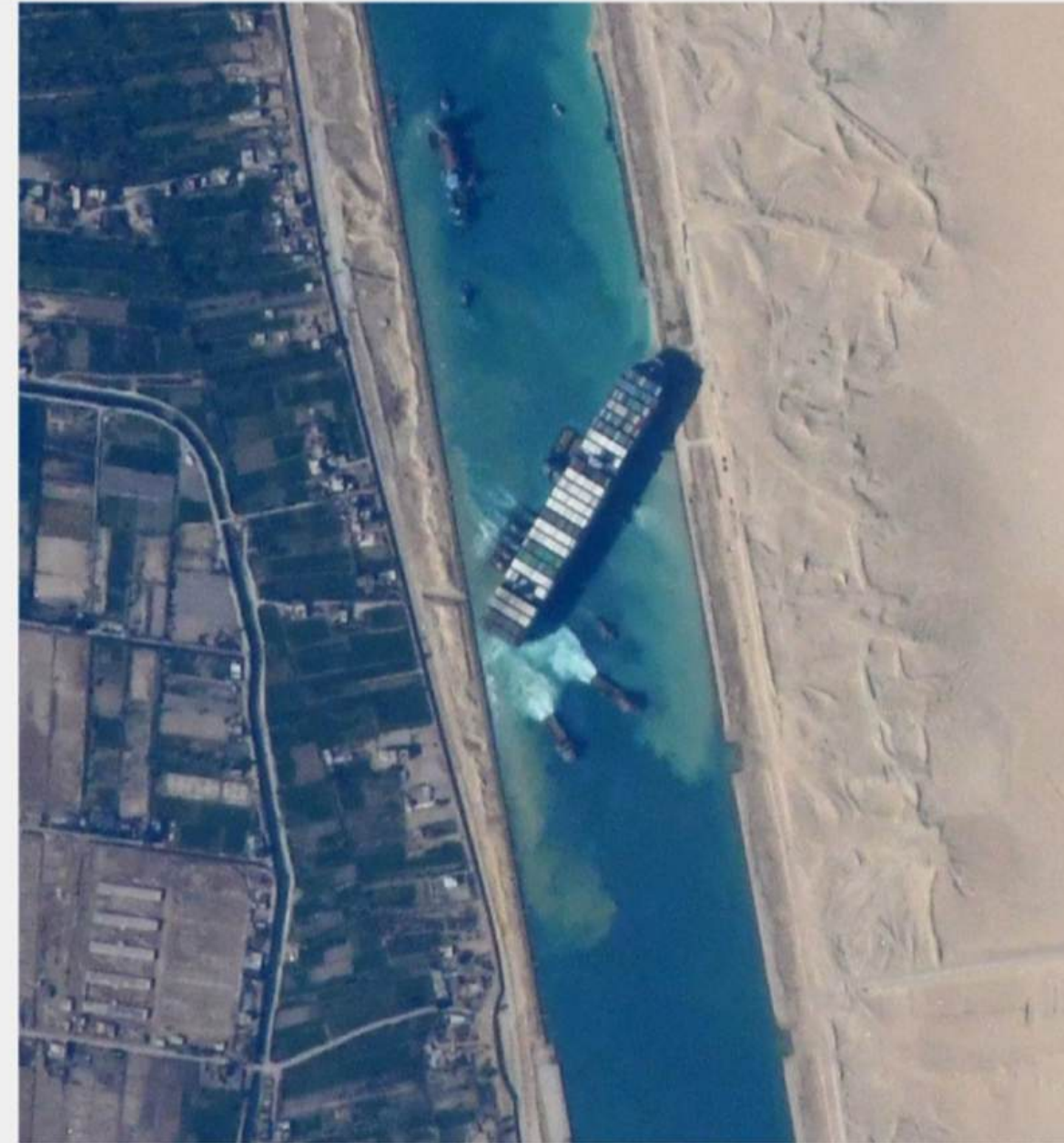
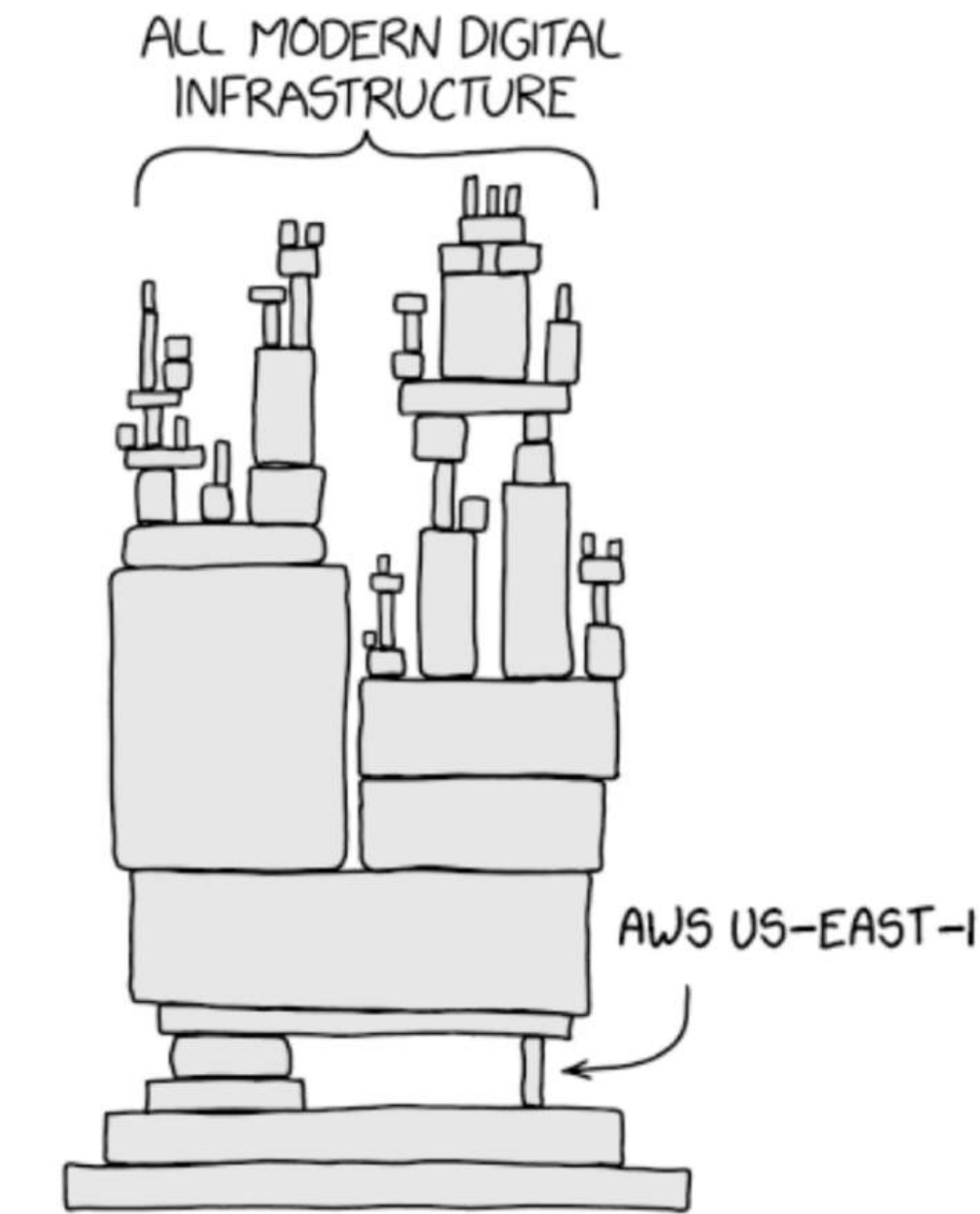
- Start with least-critical regions or edge locations
- Automated traffic analysis comparing pre/post change behavior
- Implement "single region" → "region pair" → "full deployment" progression

- **Fail-open**

- Default to safe state instead of crashing when config is corrupt
- Instead of HTTP 500, you might get degraded service. That's better than complete outage.



Incident Overview - The Basics

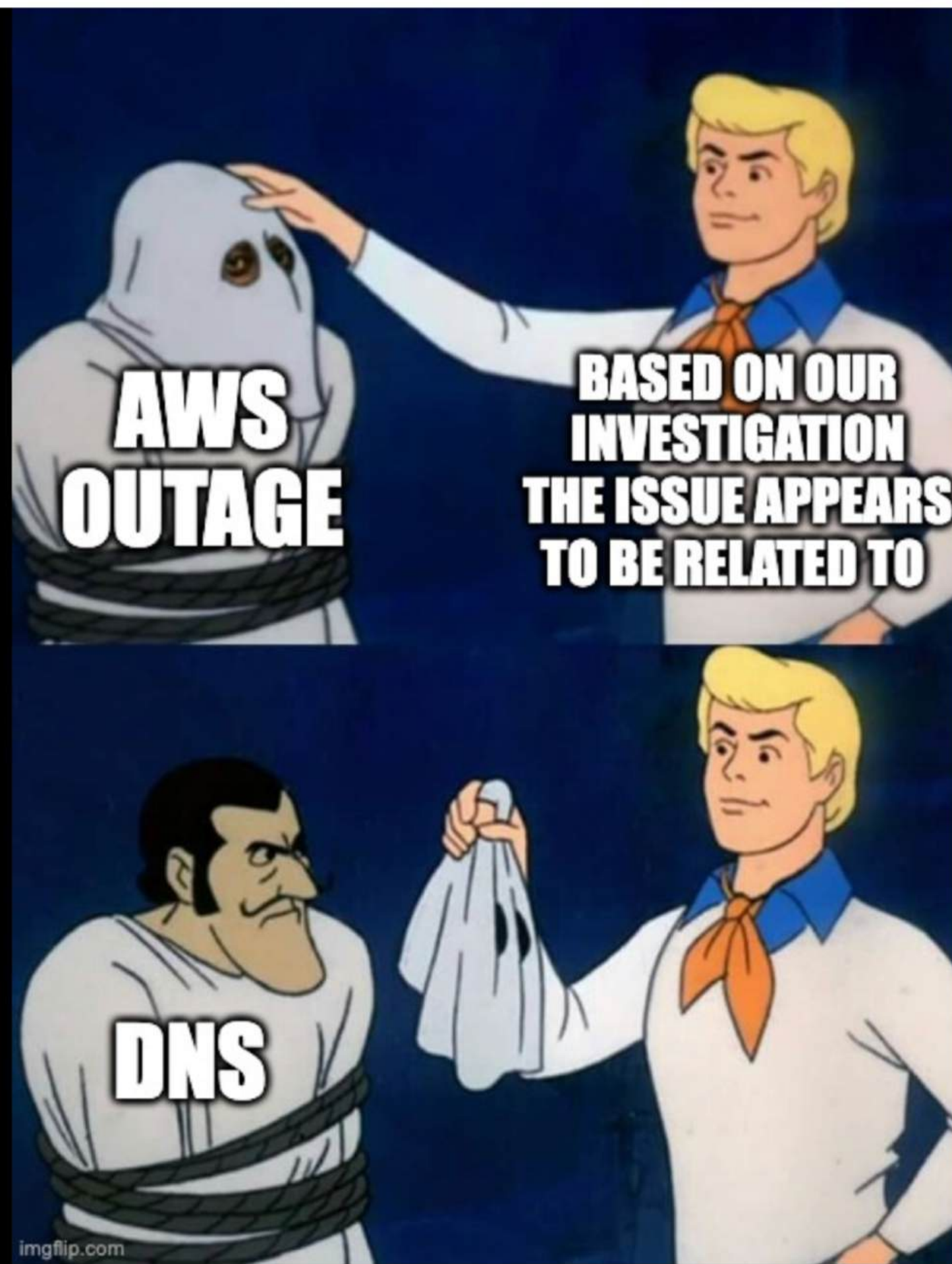


AWS US-EAST-1 (Oct 20' 25)

The Incident - Planetary Scale Disruption

- **Date & Time:** October 19, 2025, 11:49 PM PDT (October 20, 06:49 UTC)
- **Duration:** 14-15 hours of cascading failures
- **Region:** AWS US-EAST-1 (Northern Virginia)—single largest cloud region globally
- **Scope:** 140+ AWS services impacted across compute, storage, databases, networking
- **Root Cause:** Race condition in AWS internal DNS management system
- **Affected Applications:** Snapchat, Reddit, Disney+, Canva, and thousands more



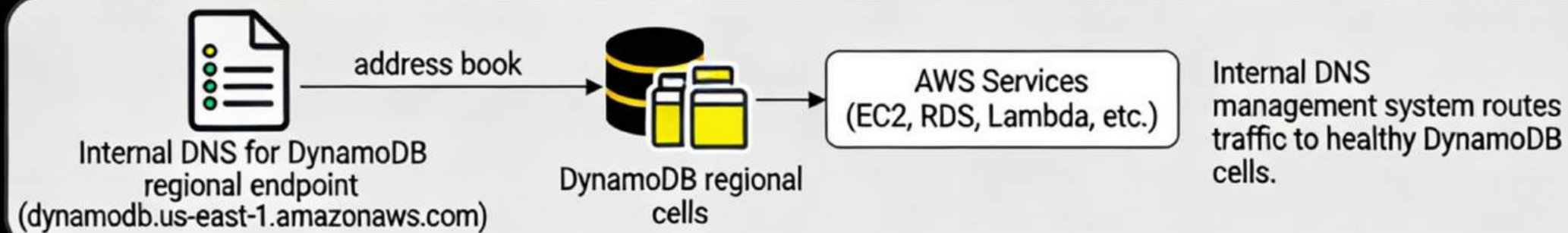


Root Cause - DNS Race Condition in Database Infrastructure

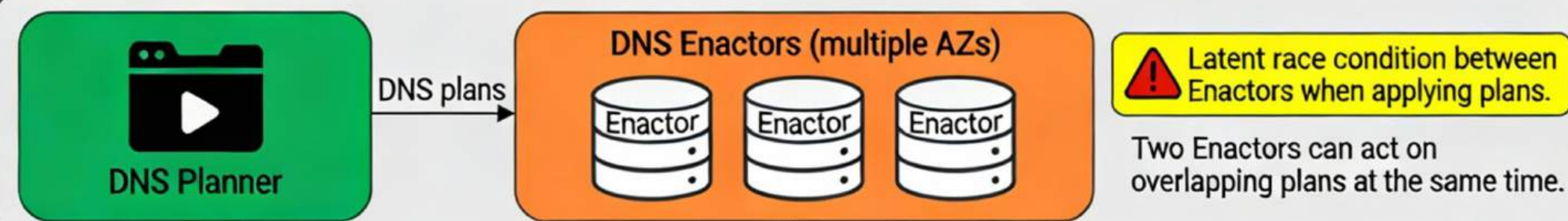
- **What Failed:** AWS internal DNS management system for RDS and database services
- **The Bug:** Race condition in automated DNS update logic between two independent system components
- **The Trigger:** Two AWS components updated DNS records simultaneously; one deleted the critical regional endpoint entry
- **Why It Matterred:** AWS services couldn't find each other; cascading failure across EC2, RDS, Lambda, etc.
- **Scope of Impact:** All services depending on AWS database infrastructure went offline
- **Detection:** Took manual intervention; AWS automated systems couldn't self-heal



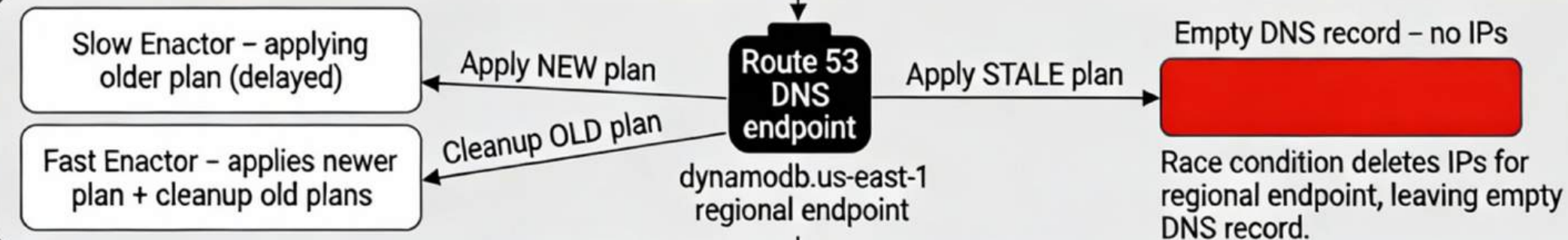
Step 1 – What Failed: Internal DNS for DynamoDB



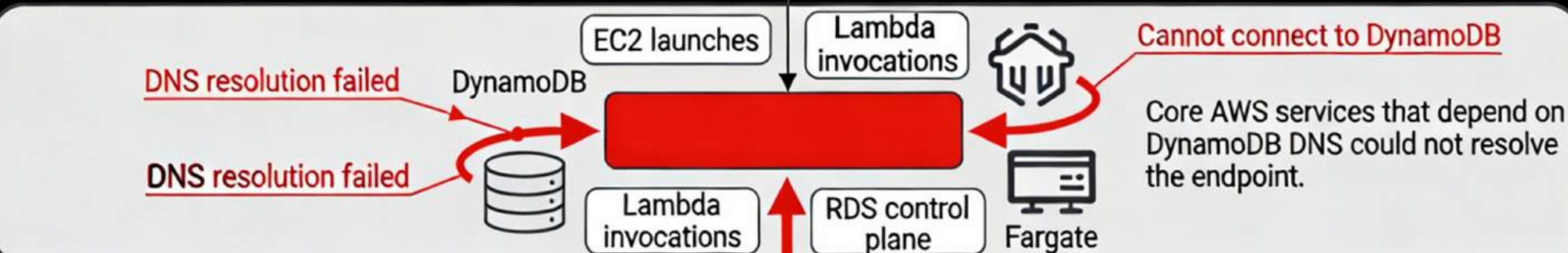
Step 2 – The Bug: Race Condition in DNS Automation



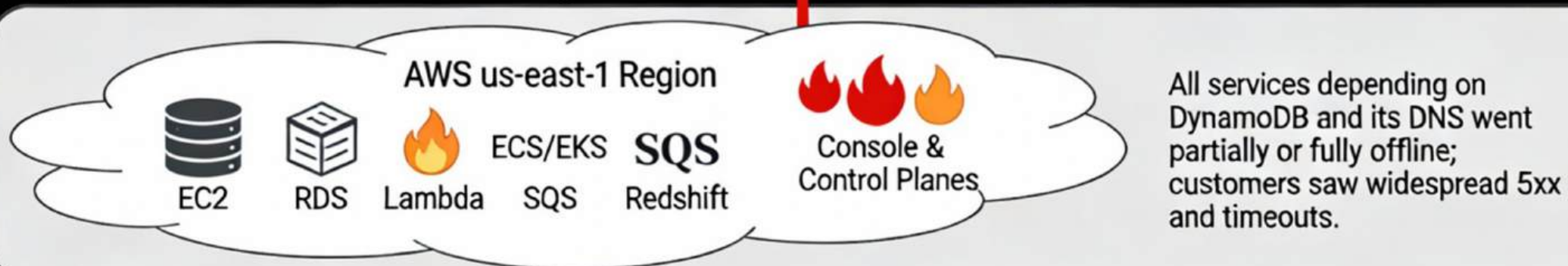
Step 3 – The Trigger: Conflicting Updates



Step 4 – Why It Mattered: Services Couldn't Find Each Other



Step 5 – Scope of Impact: Cascading Failure



AWS down

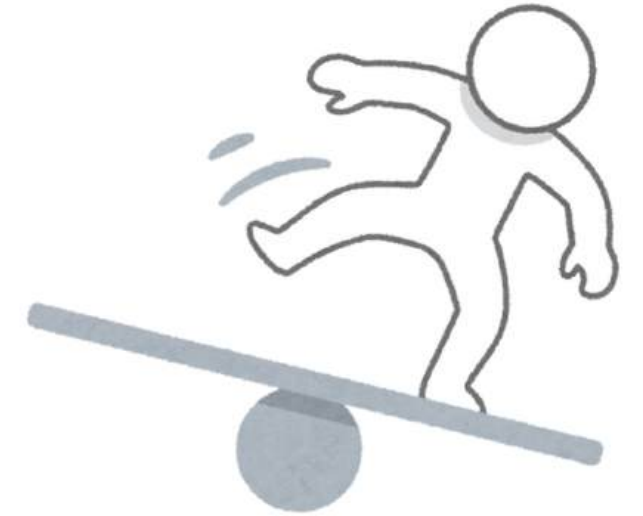
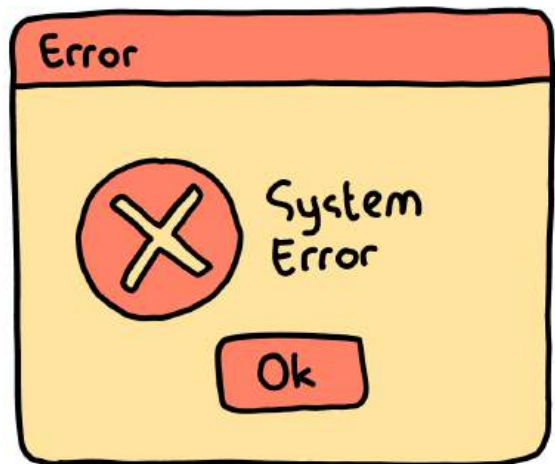
Half of the internet:



The Cascading Collapse

Chain Reaction Explained :

- DNS stops working for DynamoDB
- Other services can't find DynamoDB
- Control plane components (like IAM, EC2 workflows) failed
- Network components start flagging healthy servers as unhealthy
- Even AWS engineers have trouble accessing internal tools
- Widespread service unavailability



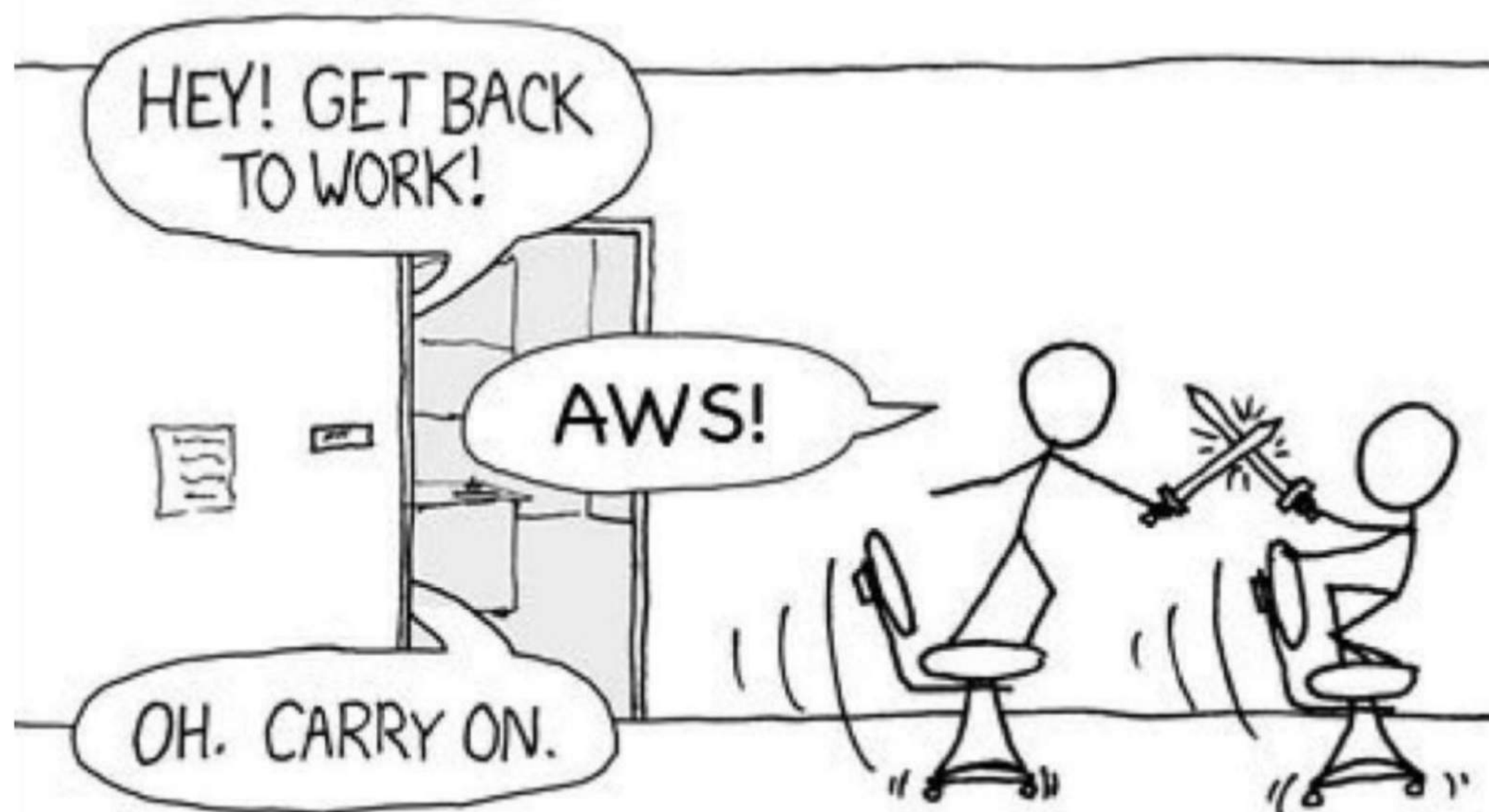
How It was fixed?

- Engineers restored the correct DNS records so services could find DynamoDB again
- They stopped faulty automation that had caused the bad DNS update
- They throttled some operations (like instance launches) to reduce pressure on recovering systems
- Services gradually came back online over many hours, with full restoration by late afternoon the next day



THE #1 PROGRAMMER EXCUSE
FOR LEGITIMATELY SLACKING OFF:

"AWS OUTAGE"



How to Prevent Similar Incidents:

- **True multi-region architecture**
 - Deploy critical services actively in multiple regions;
Eliminate SPOF
- **Eliminate hidden dependencies**



How to Prevent Similar Incidents:

- **Design for cloud provider failure**

- Implement multi-cloud or hybrid cloud for critical workloads
- Build control planes that don't depend on cloud provider control planes

- **Independent observability and communication**

- Status pages hosted outside your primary cloud provider
- Alerting systems that don't depend on the infrastructure they're monitoring



Demo



<https://github.com/CodesbyUnnati/SRE-Talk-Demo/>

The Checklist – Audit Your System Today

- **Configuration Safety:** Do configs get the similar reviews as code?
- **Urgency Process:** When urgent, do you enforce safety MORE, not less?
- **Amplification Awareness:** For each critical change, what's the blast radius?
- **Recovery Testing:** If system crashes, can you fix it remotely?
- **Dependency Mapping:** Chaos test your "independent" systems

Questions?

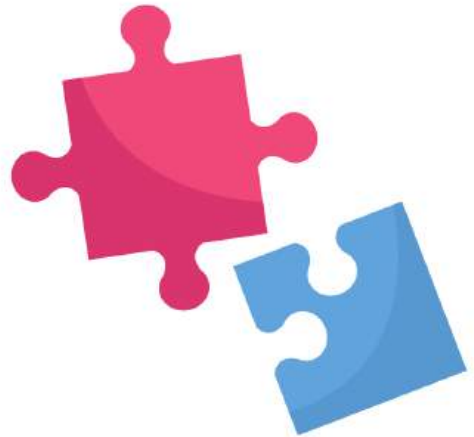


Let's Connect!



Unnati Mishra

<https://bento.me/unnati-mishra>



Thank
you

